Lab 8: Model Comparison and Regularization (pre-lab lecture)

Instructor notes. Part 1: prep. Part 2: podium and circulating cheat sheet

How to prepare (for students)

- **Big picture.** Monday compared models two ways: penalizing fit by the number of parameters (likelihood ratio tests, AIC) and by the size of the coefficients (ridge and lasso). This lab applies both to a 60-pond chytrid survey that extends Lab 7, where several pond features compete to explain where Bd is common.
- **Bring a laptop with Stan working.** We'll start by checking installs and last Wednesday's take-home WOD.
- **Review:** Monday's notes on AIC and ridge/lasso; Lab 7's logistic-binomial model and how to write its NLL; Bolker 6.6 through 6.6.2 (printed p. 267-280).
- **R skills:** `glm()` with a binomial response written as `cbind(successes, failures)`; `optim()`; `scale()`.

Overview (instructor only)

Beat	Mode	Time
0. Stan install check (before the lecture)	laptops	~10 min
1. Big picture	BOARD	~3 min
2. Catch-up from Monday (**update after Monday**)	BOARD → R	~12 min
3. Penalties as budgets	BOARD	~5 min
4. Tools for the lab	BOARD	~5 min
5. Cross-validation	BOARD	~7 min
6. Launch Act 1	BOARD	~2 min

- Lecture ~34 min; with the Stan check, Acts start ~45 min in, leaving ~2 h. Act 2 is take-home capable.
- **Don't give away:** the 2-D budget picture (Act 1 Part 2), whether to penalize β_0 or why standardize (Act 2 1a-1b), or step-by-step CV pseudocode (Act 2 1c).

Part 1: Lecture content

0. Stan check (~10 min, before lecture)

- Everyone runs the myxo model from the Week 9 take-home WOD. Pair working laptops with broken ones; collect unresolved cases for office hours rather than holding the room.

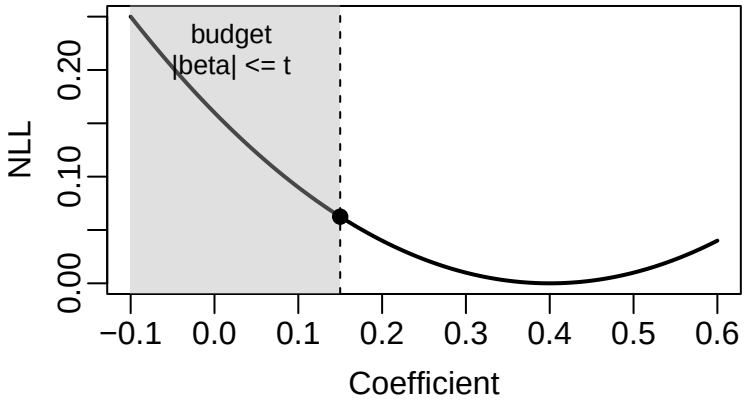
1. Big picture (~3 min) [BOARD]

- Still COMPARE (2): which features earn a place in the model?
- Today's data: 60 ponds, 8 pond features; Lab 7's model with more predictors. Temperature and elevation: r = -0.94.

2. Catch-up from Monday (~12 min) [BOARD → R]

- **Placeholder; update after Monday.** Most likely: Monday section 5 (ridge and lasso on the creepers: 1-D penalty sketch, glmnet paths, `cv.glmnet`, the shrinkage pause). Run it from the Monday section of `CODE_Week10_ModelComparison.R`.
- If Monday finished: use the time for a longer section 5 here, or start the Acts early.

3. Penalties as budgets (~5 min) [BOARD]

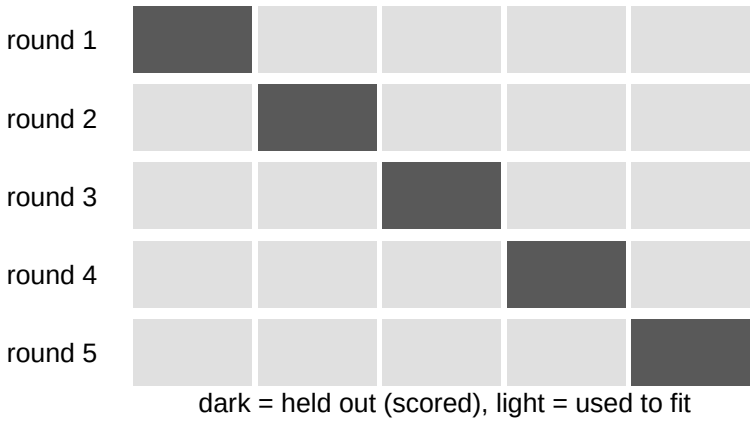


- Penalized fit NLL + $\lambda \sum \beta_j^2$ = lowest NLL inside a budget $\sum \beta_j^2 \leq t^2$; bigger λ = smaller t
- Lasso budget: $\sum |\beta_j| \leq t$
- One coefficient only (above). Stop there; Act 1 does two.

4. Tools for the lab (~5 min) [BOARD]

- `glmnet(X, y, family = "binomial", alpha = )`: `alpha = 0` ridge, `1` lasso
- Binomial counts: `y = cbind(failures, successes)` (the reverse of `glm()`'s order)
- **Sticking point:** `glmnet` scales the NLL by the sample size, so its λ is not the λ in a hand-coded penalty. Compare coefficients, not λ values.

5. Cross-validation (~7 min) [BOARD]



- Fit to 4/5 of the ponds, score the held-out 1/5 by its NLL, rotate
- Choose the λ with the best held-out score; `lambda.1se` = simplest within 1 SE
- Callback: Monday's WOD "new plot" made systematic
- Folds are random: set a seed to reproduce

6. Launch Act 1 (~2 min)

- Laptops closed; Act 2 can go home.

PART 2: PODIUM AND CIRCULATING CHEAT SHEET

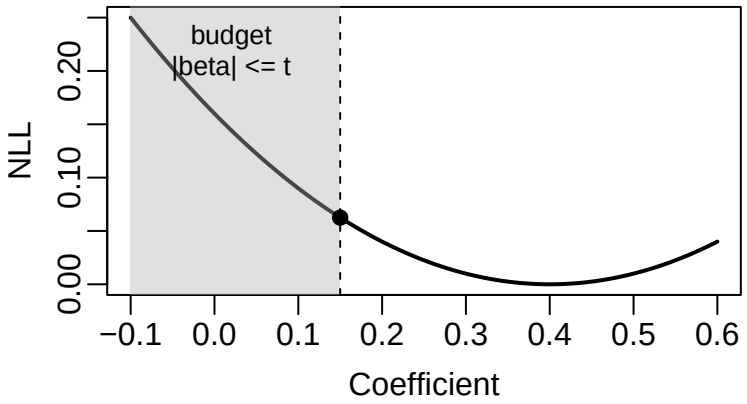
Terse on purpose. Detail: Part 1 and the two keys.

Stan check first (~10). Lecture ~34. Don't give away: 2-D budgets, β_0 penalty, standardizing, CV steps.

1-2. Big picture; catch-up (~15 min)

- COMPARE (2): which features earn a place?
- 60 ponds, 8 features; temp vs. elev r = -0.94
- Catch-up: **update after Monday**

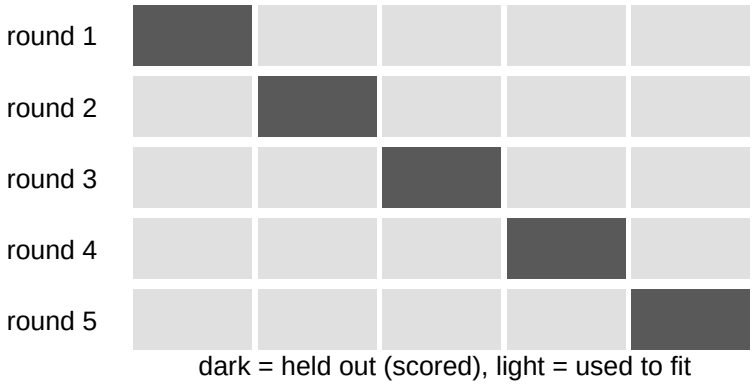
3. Budgets (~5 min) [BOARD]



- Penalty $\lambda \leftrightarrow$ budget t; bigger λ, smaller t
- Ridge $\sum \beta^2 \leq t^2$; lasso $\sum |\beta| \leq t$
- 1-D only; Act 1 does 2-D

4-5. Tools; CV (~12 min) [BOARD]

- `alpha = 0` ridge, `1` lasso; `y = cbind(failures, successes)`
- glmnet's $\lambda \neq$ hand-coded λ



- Fit 4/5, score held-out 1/5 by NLL, rotate; pick best λ
- `lambda.1se` = simplest within 1 SE; set a seed

Act-by-Act quick reference

Act 1 (paper)

- Best: Model 5 (temp + bullfrogs + canopy); Model 6 within 2 but NLL down only 0.38 (uninformative parameter)
- LRT 5 vs. 6: $D = 0.76$, 1 df; 5 vs. 7: $D = 3.98$, 5 df. Keep 5
- Temp vs. elevation can't be separated; need decoupled ponds or an experiment
- Budgets at $t = 0.4$: ridge $\approx (-0.29, 0.28)$, lasso $\approx (-0.23, 0.17)$; lasso lands mid-edge (contours parallel to the edge), so **no zero**

Act 2 (coding)

- Don't penalize β_0; standardize so one penalty fits all
- Check: temp -0.64 at $\lambda = 0$, -0.41 at $\lambda = 10$; elevation 0.16 $\rightarrow$ 0.30 (pulled toward temp)
- Lasso (`set.seed(1)`, 10 folds) keeps temp, elev, canopy, bullfrogs
- Common bugs: `cbind()` order in glmnet; penalizing β_0; forgetting to standardize; `optim()` default Nelder-Mead stalling with 9 parameters (use BFGS)